The Zero Gap of the Spike-and-Slab LASSO Global Mode: An Expanded Derivation and its Relation to the Adaptive Threshold

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This note expands the remark at the end of Section 3.1 of Ročková and George (2018), *The Spike-and-Slab LASSO*. We prove that the nonzero coordinates of the global mode are bounded away from zero (the *zero gap*), and we relate this magnitude floor δ_j to the adaptive selection threshold Δ_j of Theorem 4. Equation numbers in parentheses refer to that paper.

1 The claim

Under the conditions of Theorem 4, the global mode $\widehat{\beta}$ has a *zero gap*: every coordinate is either exactly zero or bounded away from zero, with

$$|\widehat{\beta}_j| > \delta_j \qquad \text{whenever } \widehat{\beta}_j \neq 0, \tag{1}$$

where δ_j is the unique solution of $p^{\star}_{\widehat{\theta}_j}(\delta_j) = c_+$, with $\widehat{\theta}_j = \mathrm{E}[\theta \mid \widehat{\beta}_{\setminus j}]$ and

$$c_+ = \tfrac{1}{2}\Big(1 + \sqrt{1 - 4n/(\lambda_0 - \lambda_1)^2}\Big). \tag{2}$$

This is the nonseparable analogue of the Section 2 statement $|\widehat{\beta}_j| > \delta_{c_+}$ (attributed to Lemma 4.1 of Ročková 2015). Because the coordinatewise problem freezes θ at $\widehat{\theta}_j$, the proof is identical to the separable one with $\theta = \widehat{\theta}_j$; the only new feature is that the threshold is coordinate-specific and data-driven.

2 Univariate reduction

By (3.12), with $z = z_j > 0$ fixed (WLOG, so $\widehat{\beta}_j \geq 0$) and $\theta = \widehat{\theta}_j$ held fixed,

$$\widehat{\beta}_j = \arg\max_{\beta}\ Q(\beta), \qquad Q(\beta) = -\tfrac{1}{2}(z - n\beta)^2 + n\,\rho(\beta \mid \theta), \tag{3}$$

where ρ is the separable singleton penalty (2.2),

$$\rho(\beta \mid \theta) = -\lambda_1|\beta| + \log\big[p^{\star}_{\theta}(0)/p^{\star}_{\theta}(\beta)\big]. \tag{4}$$

3 Derivatives and the curvature of the penalty

From Lemma 1, $\rho'(\beta) = -\lambda^\star_\theta(\beta)$ with $\lambda^\star_\theta(\beta) = \lambda_1 p^\star_\theta(\beta) + \lambda_0[1 - p^\star_\theta(\beta)] = \lambda_0 - (\lambda_0 - \lambda_1)\, p^\star_\theta(\beta)$. Differentiating the exponential mixing weight (2.6) gives the logistic identity

$$p^{\star\prime}_\theta(\beta) = (\lambda_0 - \lambda_1)\, p^\star_\theta(\beta)\big[1 - p^\star_\theta(\beta)\big], \tag{5}$$

so the penalty's second derivative is

$$\rho''(\beta) = (\lambda_0 - \lambda_1)^2\, p^\star_\theta(\beta)\big[1 - p^\star_\theta(\beta)\big]. \tag{6}$$

Its maximum over β is the maximal nonconcavity $\kappa = \frac{1}{4}(\lambda_0 - \lambda_1)^2$ (attained at $p^\star_\theta = \frac{1}{2}$), matching p. 433. Hence

$$Q'(\beta) = n\big[z - n\beta - \lambda^\star_\theta(\beta)\big], \qquad Q''(\beta) = n\big[\rho''(\beta) - n\big]. \tag{7}$$

4 The second-order condition forces $\widehat{\beta}$ onto the upper branch

If $\widehat{\beta} > 0$ is the global maximizer it is in particular a local maximizer, so $Q''(\widehat{\beta}) \le 0$, i.e. $\rho''(\widehat{\beta}) \le n$. With $u := p^\star_\theta(\widehat{\beta})$ and (6),

$$(\lambda_0 - \lambda_1)^2\, u(1-u) \le n \iff u(1-u) \le \frac{n}{(\lambda_0 - \lambda_1)^2} \iff u^2 - u + \frac{n}{(\lambda_0 - \lambda_1)^2} \ge 0. \tag{8}$$

The quadratic in (8) is nonnegative iff

$$u \le c_- \quad \text{or} \quad u \ge c_+, \qquad c_\mp = \tfrac{1}{2}\Big(1 \mp \sqrt{1 - 4n/(\lambda_0 - \lambda_1)^2}\Big), \tag{9}$$

with real, distinct roots precisely when $(\lambda_0 - \lambda_1)^2 > 4n$. Thus no local maximum can lie in the curvature band $c_- < p^\star_\theta(\beta) < c_+$, where Q is strictly convex ($Q'' > 0$). Consequently Q is concave–convex–concave in β, and the only nonzero maximizer candidates are a point on the *lower* branch ($p^\star_\theta \le c_-$, small β) or the *upper* branch ($p^\star_\theta \ge c_+$, larger β).

Excluding the lower branch. Write $h(\beta) := Q'(\beta)/n = z - n\beta - \lambda^\star_\theta(\beta)$, so by (7) $h'(\beta) = \rho''(\beta) - n$ shares the sign pattern of (9): h decreases on $(0, \delta_{c_-})$, increases across the convex band, and decreases again for $\beta > \delta_{c_+}$. Near the origin the spike slope dominates, $\lambda^\star_\theta(0) \approx \lambda_0$, so Q falls away from 0 across the first concave piece and any lower-branch stationary point is dominated by the mode at 0. The condition $g_\theta(0) > 0$ — equivalently $\lambda^\star_\theta(0) > \sqrt{2n \log[1/p^\star_\theta(0)]} + \lambda_1$ (p. 435) — is exactly what guarantees this domination, so the only nonzero value that can beat $Q(0)$ lies on the upper decreasing branch. (This is the content of Lemma 4.1 of Ročková 2015; the second-order argument above is the crux.) Hence a nonzero global mode satisfies $p^\star_\theta(\widehat{\beta}) \ge c_+$.

5 Converting the branch condition into the magnitude floor

Since $p^\star_\theta$ is strictly increasing, $p^\star_\theta(\widehat{\beta}) \ge c_+ \iff \widehat{\beta} \ge \delta_j$, where δ_j solves $p^\star_{\widehat{\theta}_j}(\delta_j) = c_+$. Solving (2.6) explicitly,

$$c_+ = \Big[1 + \tfrac{\lambda_0}{\lambda_1} \tfrac{1-\theta}{\theta} e^{-\delta_j(\lambda_0 - \lambda_1)}\Big]^{-1},$$

gives

$$\boxed{\delta_j = \frac{1}{(\lambda_0 - \lambda_1)} \log\left[\frac{1 - \widehat{\theta}_j}{\widehat{\theta}_j} \cdot \frac{\lambda_0}{\lambda_1} \cdot \frac{c_+}{1 - c_+}\right].} \tag{10}$$

With a single fixed θ this is exactly δ_{c_+} of Section 2; with $\widehat{\theta}_j = \mathrm{E}[\theta \mid \widehat{\beta}_{\setminus j}]$ it becomes the coordinate-specific gap. Because $z > \Delta_j$ at selection places the stationary point strictly interior to the upper branch, the inequality is strict, establishing $|\widehat{\beta}_j| > \delta_j$ and hence (1). $\square$

6 Relation to the adaptive threshold Δ_j of Theorem 4

The two thresholds answer different questions and arise from different parts of the same geometry.

Δ_j is a threshold on the data z_j — it governs *selection*. By Theorem 3, $\widehat{\beta}_j = 0 \iff |z_j| \leq \Delta_j$, and Theorem 4 bounds it by $\Delta_j^L < \Delta_j < \Delta_j^U$ with $\Delta_j^U = \sqrt{2n \log[1/p^\star_{\widehat{\theta}_j}(0)]} + \lambda_1$. It is a *global* comparison — the value of z_j at which $Q(\widehat{\beta})$ overtakes $Q(0)$ — which is why its scale is $\sqrt{2n \log[1/p^\star_{\widehat{\theta}_j}(0)]}$ and it is driven by $p^\star_{\widehat{\theta}_j}(0)$, the shrinkage at the origin.

δ_j is a threshold on the estimate $\widehat{\beta}_j$ — it governs the *magnitude* of whatever survives selection. It is a *local* condition: the boundary $Q'' = 0$ of the concave region, which is why its scale is set by c_+, i.e. by $(\lambda_0 - \lambda_1)^2$ versus n, rather than by $p^\star_{\widehat{\theta}_j}(0)$.

They are tied together in three ways.

1. **Same adaptive driver.** Both inherit coordinate-adaptivity from the single learned quantity $\widehat{\theta}_j = \mathrm{E}[\theta \mid \widehat{\beta}_{\setminus j}]$. As more sparsity is detected in $\widehat{\beta}_{\setminus j}$, $\widehat{\theta}_j$ falls; this raises Δ_j^U (a smaller $p^\star_{\widehat{\theta}_j}(0)$, harder to enter the model) while simultaneously raising δ_j (through the $(1 - \widehat{\theta}_j)/\widehat{\theta}_j$ factor in (10)). The Scott–Berger multiplicity adjustment thus moves the selection threshold and the gap *in concert*, coordinate by coordinate — the nonseparable counterpart of the single fixed pair (Δ^U, δ_{c_+}) in Section 2.

2. **The selection threshold *creates* the gap.** At the exact selection boundary $z_j = \Delta_j$ the competing nonzero maximizer is already on the upper branch ($p^\star_\theta \geq c_+$). So as z_j crosses Δ_j the estimate jumps discontinuously from 0 to a value already exceeding δ_j — the hard-thresholding face of the "blend of soft- and hard-thresholding" in Theorem 3. There is no continuum of tiny nonzero estimates; that is precisely the zero gap.

3. **Common large-λ_0 limit.** As $(\lambda_0 - \lambda_1) \to \infty$, $c_+ \to 1$ and $1 - c_+ = c_- \to n/(\lambda_0 - \lambda_1)^2$, so $c_+/(1 - c_+) \approx (\lambda_0 - \lambda_1)^2/n$ and

$$\delta_j \approx \frac{1}{(\lambda_0 - \lambda_1)} \log\left[\frac{1 - \widehat{\theta}_j}{\widehat{\theta}_j} \cdot \frac{\lambda_0}{\lambda_1} \cdot \frac{(\lambda_0 - \lambda_1)^2}{n}\right] \longrightarrow 0.$$

   The gap floor vanishes while $\Delta_j \to \Delta_j^U$, so the estimator approaches the ℓ_0/hard-threshold limit in which selection is controlled entirely by Δ_j and surviving coefficients are essentially unbiased — matching the Section 2 remark that $\Delta \to \Delta^U$ for large $(\lambda_0 - \lambda_1)$.

In short: Δ_j (from the *global* value comparison, scale $\sqrt{2n \log[1/p^\star_{\widehat{\theta}_j}(0)]}$) decides *whether* a coordinate is nonzero; δ_j (from the *local* curvature boundary, scale set by c_+) decides *how small a nonzero coordinate can be.* Both are made coordinate-adaptive by $\widehat{\theta}_j$, and they meet at the selection boundary, which is what makes the gap appear.

A The penalty derivative: a detailed derivation of Lemma 1

The proof of Lemma 1 opens from the identity

$$\frac{\partial\,\mathrm{pen}_S(\boldsymbol{\beta}\mid\theta)}{\partial|\beta_j|} = p^\star_\theta(\beta_j)\,\frac{\partial\log\psi_1(\beta_j)}{\partial|\beta_j|} + \big[1-p^\star_\theta(\beta_j)\big]\,\frac{\partial\log\psi_0(\beta_j)}{\partial|\beta_j|}. \tag{11}$$

It is worth noting that (**??**) is *not* the derivative of the regrouped singleton (2.2); it is the derivative of the original mixture form (2.1), and (2.2) is an algebraically equivalent regrouping of the same object. We record both routes.

Setup. The two Laplace components (1.3) are

$$\psi_1(\beta) = \tfrac{\lambda_1}{2}e^{-\lambda_1|\beta|}, \qquad \psi_0(\beta) = \tfrac{\lambda_0}{2}e^{-\lambda_0|\beta|},$$

each depending on β only through $t=|\beta|$, so that

$$\frac{\partial\log\psi_1}{\partial|\beta|} = -\lambda_1, \qquad \frac{\partial\log\psi_0}{\partial|\beta|} = -\lambda_0. \tag{12}$$

The per-coordinate term of the penalty, *before* the rewrite into (2.2), is the mixture form (2.1),

$$s(\beta_j) = \log\big[\theta\psi_1(\beta_j) + (1-\theta)\psi_0(\beta_j)\big] - \log\big[\theta\psi_1(0) + (1-\theta)\psi_0(0)\big],$$

whose second bracket is constant in β_j.

Where (??) comes from. Write the mixture density $f(\beta) := \theta\psi_1(\beta) + (1-\theta)\psi_0(\beta)$. Then, with $'$ denoting $\partial/\partial|\beta|$,

$$\frac{\partial s}{\partial|\beta_j|} = \frac{\partial\log f}{\partial|\beta_j|} = \frac{f'}{f} = \frac{\theta\psi_1'(\beta_j) + (1-\theta)\psi_0'(\beta_j)}{f}.$$

Applying the standard log-mixture derivative identity — multiply and divide each term by its own component density —

$$\frac{f'}{f} = \frac{\theta\psi_1}{f}\cdot\frac{\psi_1'}{\psi_1} + \frac{(1-\theta)\psi_0}{f}\cdot\frac{\psi_0'}{\psi_0} = \frac{\theta\psi_1}{f}\,\frac{\partial\log\psi_1}{\partial|\beta_j|} + \frac{(1-\theta)\psi_0}{f}\,\frac{\partial\log\psi_0}{\partial|\beta_j|}.$$

The two weights are exactly the conditional mixing probability (2.4) and its complement,

$$\frac{\theta\psi_1(\beta_j)}{f} = p^\star_\theta(\beta_j), \qquad \frac{(1-\theta)\psi_0(\beta_j)}{f} = 1 - p^\star_\theta(\beta_j), \tag{13}$$

which yields (**??**). Thus the identity is the score of a log-mixture, equal to the *responsibility-weighted average of the component scores*. Here $p^\star_\theta(\beta_j)$ is not incidental: it is the posterior probability $\mathrm{P}(\gamma_j = 1 \mid \beta_j, \theta)$ of (2.6), i.e. the responsibility that the slab ψ_1 generated β_j — the same weight that appears in the E-step of EM — which is why it drops out immediately. Substituting (**??**),

$$\frac{\partial\,\mathrm{pen}_S(\boldsymbol{\beta}\mid\theta)}{\partial|\beta_j|} = -\lambda_1 p^\star_\theta(\beta_j) - \lambda_0\big[1-p^\star_\theta(\beta_j)\big] = -\lambda^\star_\theta(\beta_j),$$

which is (2.7). □

Reconciliation with the regrouped form (2.2). Differentiating (2.2) directly must give the same answer; it merely hides the mixture step inside the $p^\star$ term. From $\rho(\beta_j \mid \theta) = -\lambda_1|\beta_j| + \log p^\star_\theta(0) - \log p^\star_\theta(\beta_j)$,

$$\frac{\partial\rho}{\partial|\beta_j|} = -\lambda_1 - \frac{\partial \log p^\star_\theta(\beta_j)}{\partial|\beta_j|}.$$

Evaluating the last derivative still requires the mixture: since $\log p^\star_\theta = \log\theta + \log\psi_1 - \log f$,

$$\frac{\partial \log p^\star_\theta}{\partial|\beta_j|} = \frac{\partial \log\psi_1}{\partial|\beta_j|} - \frac{\partial \log f}{\partial|\beta_j|} = -\lambda_1 - \big(-\lambda^\star_\theta(\beta_j)\big) = \lambda^\star_\theta(\beta_j) - \lambda_1,$$

using $\partial \log f/\partial|\beta_j| = -\lambda^\star_\theta(\beta_j)$ from above. Hence

$$\frac{\partial\rho}{\partial|\beta_j|} = -\lambda_1 - \big(\lambda^\star_\theta(\beta_j) - \lambda_1\big) = -\lambda^\star_\theta(\beta_j),$$

in agreement. The two forms coincide because (2.2) is the "LASSO piece plus $\log p^\star$ correction" regrouping of (2.1) (the passage from 2.1 to 2.3); but differentiating (2.2) directly forces one to compute $\partial \log p^\star_\theta/\partial|\beta_j|$, which reintroduces the mixture. Differentiating the mixture form first lets the log-derivative identity (**??**) collapse everything to the $p^\star$-weighted average in a single line.